

Technical Report: Student Performance Prediction

InspiHER{Tech} V4.0 - MLArena Post Workshop Task
Team Hacktronix

1. Objective

The main objective of this project is to develop a Machine Learning classification model capable of predicting student academic results (Pass or Fail) using academic and behavioral features such as attendance, internal test scores, assignment performance, and study habits. The project demonstrates the practical use of supervised learning algorithms in educational data analysis and student performance prediction.

2. Project Overview

Developed as part of the InspiHER{Tech} V4.0 Machine Learning Arena challenge, this project covers the end-to-end ML workflow, from raw data acquisition to model deployment strategies. The focus remains on interpretability and predictive accuracy.

3. Methodology

The project followed a structured Machine Learning pipeline to ensure reliable model development and evaluation.

3.1 Dataset Features

Feature	Description
Attendance (%)	Percentage of student attendance
Internal Test 1 and 2	Marks obtained in first and second internal examination
Assignment Score	Assignment marks obtained by students
Daily Study Hours	Average daily study time
Final Exam Marks	Final examination marks
Result (Target)	Student outcome (Pass or Fail)

3.2 Feature Engineering

Two additional features were created to improve model performance and better represent student academic behavior.

Engineered Feature	Description
CA_Average	Average continuous assessment score calculated using internal tests and assignments
Study_Efficiency	Combined measure of attendance and daily study hours

3.3 Data Preprocessing & Workflow

The following preprocessing steps were applied before model training:

- Data loading using Pandas
- Checking and handling missing values
- Domain-based outlier detection using valid academic score ranges
- Statistical outlier analysis using the IQR method
- Label Encoding for converting Pass/Fail into numerical values
- Feature selection by removing Student_ID
- Splitting the dataset into 70% training data and 30% testing data
- Class imbalance analysis on the training dataset
- SMOTE balancing technique was tested for comparison purposes

4. Machine Learning Algorithms

Several classification algorithms were trained and evaluated:

- Logistic Regression ,Decision Tree Classifier, Random Forest Classifier, K-Nearest Neighbors (KNN), XGBoost Classifier

Both SMOTE and non-SMOTE datasets were tested to compare model performance and identify the most reliable approach.

5. Model Performance

The models were evaluated using Accuracy, Precision, Recall, and F1-Score. After comparison, Logistic Regression trained on the original non-SMOTE dataset achieved the best performance.

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	94.83%	96.55%	97.79%	97.17%

The confusion matrix showed that the model correctly predicted most student outcomes with only a few misclassifications.

7. Future Improvements

- Use larger and more diverse educational datasets
- Include additional behavioral and demographic features
- Apply hyperparameter tuning for model optimization
- Deploy the model using Streamlit or Flask for real-time prediction
- Explore deep learning approaches for advanced prediction accuracy

8. Conclusion

This project successfully demonstrated the use of Machine Learning techniques for predicting student academic performance. Multiple classification algorithms were tested and compared using several evaluation metrics. Logistic Regression achieved the best performance with an accuracy of 94.83% and an F1-score of 97.17%. The analysis shows that academic behavior and continuous assessment features are highly useful in predicting student outcomes. Such predictive systems can support educational institutions in identifying students who may require additional academic assistance.